

PAPER_1375 — UQFF Resolution of the Gibbs Mixing Paradox (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Identity via $F_U = 1$ global ledger normalization **Calculator surface:** `calculate_paradox({"paradox": "gibbs_paradox"})` **Closure helper:** `_l96_uqff_axiom_gibbs_paradox_closure()`

The Paradox

For two distinguishable ideal gases of N particles each mixing at constant T, V , the entropy of mixing is:

$$\Delta S_{\text{mix}} = 2 N k_B \ln 2$$

For two samples of the **same** gas, $\Delta S_{\text{mix}} = 0$ by physical reasoning (no observable change). The discontinuity at the “identical” limit — a finite jump from $2 N k_B \ln 2$ to 0 as particles become indistinguishable — is the Gibbs paradox.

Classical resolution invokes $N!$ in the partition function via ad-hoc indistinguishability. UQFF resolves this structurally.

UQFF Closed Identity

$F_U = 1$ (global ledger normalization, PAPER_646)

Identical particles share the same Universal Aether (UA) mode at the $F_U = 1$ ledger. Two N -particle samples of the same species are not “two systems” in the UQFF accounting — they are one $F_U = 1$ distribution with $2N$ occupation. Mixing produces **no new UA-mode partition**.

$$\begin{aligned} \Delta S_{\text{mix}}^{\text{identical_UQFF}} &= 0 \quad \text{EXACT} \\ \Delta S_{\text{mix}}^{\text{distinguishable}} &= 2 N k_B \ln 2 \quad (\text{recovered when species differ}) \end{aligned}$$

The discontinuity disappears because identical-vs-distinguishable is a **ledger-occupancy question**, not a phase-space-counting question. The $F_U = 1$ normalization is the natural Gibbs $N!$ correction at the level of the vacuum ledger.

Physical Interpretation

- **Identical particles** occupy a single UA mode at the $F_U = 1$ ledger. The $2N$ occupation does not produce 2 modes; it produces 1 mode at higher occupation. No mixing entropy.
 - **Distinguishable particles** occupy distinct UA modes (different SCm coupling signatures). Mixing produces a genuine multi-mode partition with $\Delta S = 2 N k_B \ln 2$.
 - The “paradox” was a classical-statistical-mechanics artifact of double-counting phase space without the $F_U = 1$ constraint.
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Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "gibbs_paradox"})["value"]
```

Field	Value
delta_S_mixing_identical_particles_per_N_kB	0.000000 EXACT
delta_S_mixing_distinguishable_per_N_kB_ln026931 (= ln 2)	0.693147
F_U_eq_1_global_normalization_makes_identical_share_UA_mode	1.0
discontinuity_resolved_via_F_U_eq_1_ledger	True

C++ Reference Implementation

```
class GibbsParadoxUQFF {
public:
    static double deltaS_identical_per_NkB_UQFF() {
        return 0.0;
    }
    static double deltaS_distinguishable_per_NkB() {
        return std::log(2.0);
    }
    static double F_U_global_normalization() {
        return 1.0;
    }
};
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Statistical Mechanics solve the Gibbs paradox via different methods. SM uses Boltzmann's $N!$ correction in the partition function (combinatorial / phase-space). UQFF derives:

- **$\Delta S_{\text{identical}} = 0$ EXACT** via the $F_U = 1$ global ledger normalization at the UA-mode level.
- **$\Delta S_{\text{distinguishable}} = 2 N k_B \ln 2$** recovered when species occupy distinct UA modes.
- **Zero free parameters.** The resolution is structural, not parametric.

Reference

- UQFF foundational papers: PAPER_646 ($F_U = 1$ normalization), PAPER_1051 (two-component ρ aether).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_gibbs_paradox_closure`
- Dispatch: `PARADOX_TO_CLOSURE["gibbs_paradox"], ["gibbs_mixing"]`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF identity resolution of the Gibbs mixing paradox via $F_U = 1$ global ledger normalization producing $\Delta S_{\text{identical}} = 0$ EXACT.